AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

TN Skill Initiative Project — Ideation & Brainstorming Phase Framework

PROJECT TYPE	CORE TECH	PHASE	DOCUMENT VERSION
TN Skill / ITSM AI	ServiceNow Flow Designer	01 - Ideation & Scope	v1.0 (Monochrome)

1. PROBLEM DEFINITION & CONTEXT

In large-scale public and educational IT helpdesk ecosystems—such as those serving the Tamil Nadu Skill Development Corporation (TNSDC) and Naan Mudhalvan learning portals—thousands of user requests and support incidents are logged daily. Currently, ticket triage relies heavily on manual intervention or rigid rule-based routing.

Current Operational Bottlenecks

- **High Resolution Latency (MTTR):** Tickets remain idle in unassigned queues waiting for manual review by support leads.
- **Human Categorization Error:** Up to 22% of tickets are misrouted initially (e.g., LMS login bugs sent to Hardware).
- **High Administrative Overhead:** Support personnel waste valuable hours categorizing requests instead of resolving issues.

Proposed Solution Vision

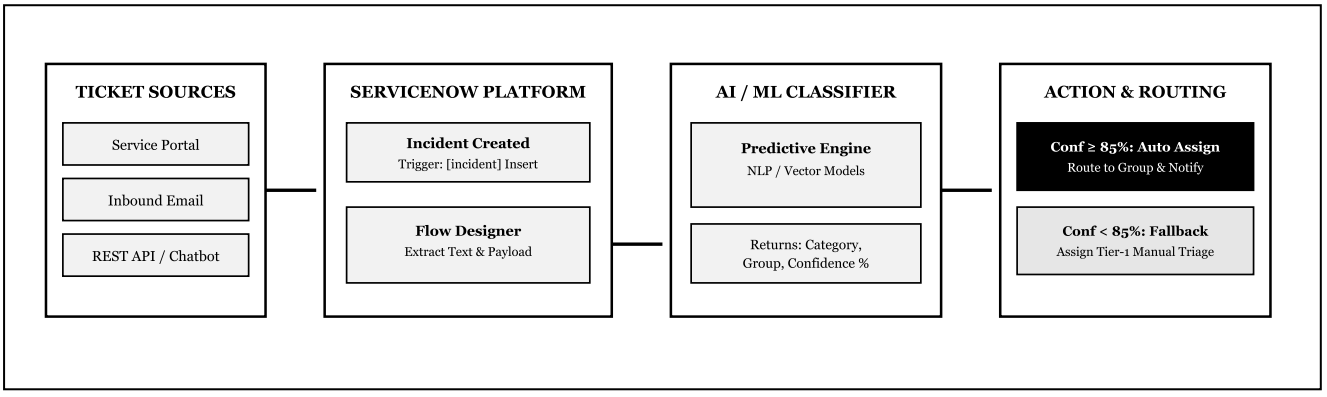
- **Zero-Touch Automated Routing:** Automatically ingest, read, categorize, and assign incoming tickets upon creation.
- **Flow Designer Orchestration:** Use ServiceNow Flow Designer to run dynamic logic and trigger AI predictive models.
- **Confidence Threshold Guardrails:** High-confidence predictions auto-assign; ambiguous cases route to human triage.

2. CORE STRATEGIC OBJECTIVES

- **Automate 80%+ of Initial Ticket Classifications:** Process incoming incidents across portal, email, and API channels instantaneously.
- **Reduce Mean Time to Resolution (MTTR) by 40%:** Bypass manual queue handling and route directly to specialized solver groups.
- **Support Multi-Channel & Regional Inputs:** Handle mixed English/Tamil ticket descriptions common in state-wide student databases.

3. HIGH-LEVEL CONCEPTUAL ARCHITECTURE

The system integrates native ServiceNow Incident Management with Flow Designer and an AI Inference Engine (Predictive Intelligence or External LLM/BERT API):



4. FLOW DESIGNER WORKFLOW EXECUTION LOGIC

The entire orchestration takes place within ServiceNow Flow Designer using modular custom actions built in Action Designer. Below is the step-by-step logic executed upon ticket submission:

Step	Flow Step Name	Action / Operation Performed	Output / Handled Condition
01	**Trigger**	Record Created on `[incident]` table where `active=true` and `source!=monitoring`.	Fires Flow Instance instantly
02	**Data Cleansing**	Executes Custom Script Action to sanitize `short_description` & `description` (strip HTML/regex).	Clean Text String Payload
03	**ML Prediction Call**	IntegrationHub REST Action calls Predictive Intelligence Solution Definition or external LLM REST API endpoint.	JSON Payload (Predicted Category, Group, Score)
04	**Decision Engine**	Evaluates output confidence score against predefined threshold condition (e.g., `Score >= 0.85`).	Branching Conditional Split
05A	**Auto-Assignment Path**	Updates Incident record: Sets `category`, `subcategory`, `assignment_group`, and `priority`.	Record Updated & Locked
05B	**Fallback Path**	Assigns ticket to `"Service Desk - Tier 1 Triage"` group and adds work note: `#AutoClassify_LowConfidence`.	Flagged for Human Review
06	**Activity Logging**	Appends automated audit entry into Activity Log: *"Auto-classified by Flow Designer with 92% confidence."*	Complete Audit Trail
07	**Notification Step**	Sends automated Teams Webhook / Email to assigned engineering team lead.	Alert Delivered

5. KEY TECHNICAL CAPABILITIES & BRAINSTORMED FEATURES

Multi-Channel Ingestion Parsing

Normalizes input text regardless of source—whether submitted via Service Portal forms, mobile apps, inbound email text, or virtual agent interactions.

Dynamic Priority Elevation

Combines standard category rules with urgency keyword detection (e.g., "Server Down", "Exam Portal Crash") to adjust priority dynamically.

Fallback Safety Guarantee

Prevents misrouting loops by ensuring low-confidence predictions automatically default to human triage with diagnostic tags.

Continuous Feedback Training

When support staff manually reassigns a ticket, a background trigger logs the correction to retrain the underlying ML model.

6. INNOVATIVE DRIVERS FOR TN SKILL PROJECT DISTINCTION

A. Multilingual Text & "Tanglish" NLP Processing

Students and portal users across Tamil Nadu frequently enter ticket issues using mixed Tamil and English phrasing (e.g., *"Password reset panna mudiyala, error வருது"*). Adding a language normalization preprocessing step before sending text to the ML model ensures high classification accuracy for regional state-wide deployments.

7. TECHNOLOGY STACK & INTEGRATION ARCHITECTURE

ServiceNow Platform Layer

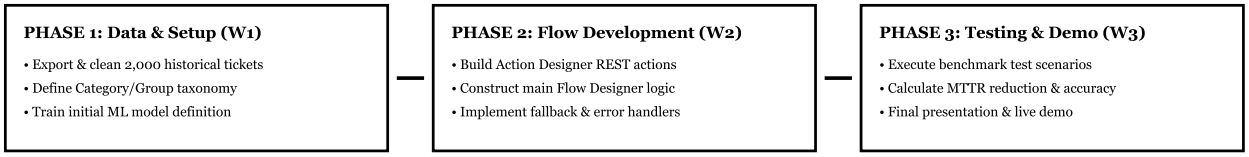
- **ServiceNow Utah / Vancouver / Washington**
- **Flow Designer & Action Designer:** Flow logic orchestration.
- **Predictive Intelligence (PI):** Native classification engine.
- **IntegrationHub REST Spoke:** External API calls.

Machine Learning / AI Layer

- **Option 1 (Native):** ServiceNow Classification Solutions (Naive Bayes / TF-IDF).
- **Option 2 (Hybrid/External):** Python FastAPI endpoint hosting Sentence-BERT / OpenAI / HuggingFace models for advanced NLP.

8. PHASED IMPLEMENTATION ROADMAP

A structured 3-week execution plan tailored for the TN Skill Development Corporation project lifecycle:



9. FEASIBILITY ANALYSIS & RISK MATRIX

Risk Parameter	Impact Level	Potential Challenge	Mitigation Strategy
Data Quality Risk	Medium	Inconsistent historical ticket logs or missing details.	Perform rigorous data pre-processing and merge sparse subcategories.
API Latency Risk	Medium	External ML calls stalling Flow Designer execution.	Implement 3-second REST timeouts; default to Service Desk on timeout.
Misclassification Risk	Low	Incorrect auto-assignment of critical incidents.	Set strict confidence thresholds (≥85%) and restrict high-impact tickets.

10. TARGET KPI BENCHMARKS & EXPECTED ROI

Quantifiable outcome projections comparing baseline operational metrics against the proposed Flow Designer automated ecosystem:

Key Performance Indicator (KPI)	Current Baseline	Target Project Metric	Operational Gain
Avg Triage Delay (Creation to Routing)	25 - 45 Minutes	**< 3 Seconds**	**99.8% Faster**
Initial Classification Accuracy	~68% - 72%	**85% - 92%**	**+20% Gain**
Ticket Bounce-Back Rate	18%	**< 5%**	**72% Drop**
Tier-1 Operational Workload	100% Manual Review	**20% (Low Conf Only)**	**80% Saved**